1. (a) 1011 (b) 0110 (c) 1110

3. Consider the (3, 4) parity check code. For each of the received words, determine whether an error will be detected.

(a) 0100 (b) 1100

6. Consider the $(m, 3m)$ encoding function of Example 3, where $m = 4$. For each of the received words, determine whether an error will be detected.

(a) 011010011111 (b) 110110010110

9. Find the distance between x and y.

(a) $x = 1100010$, $y = 1010001$

(b) $x = 0100110$, $y = 0110010$

11.1.

P414-1.. (a) 3 (b) 2 (c) 3

10

3. (a) Y (b) N.

6. (a) Y (b) Y.

9. (a) 4 (b) 2

P412- 13. 2.

25.

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$e(00)=00000$ $e(01)=01011$ $e(10)=10011$ $e(11)=11000$

26.

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

25. Let

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Determine the (2, 5) group code function $e_H : B^2 \to B^5$.

26. Let

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Determine the (3, 6) group code $e_H : B^3 \to B^6$.

11.1. 10

P414-1.. (a) 3 (b) 2 (c) 3

1 3. (a) Y (b) N.

6. (a) Y (b) Y.

9. (a) 4 (b) 2

P412- 13. 2.

25.

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

$e(00)=00000$ $e(01)=01011$ $e(10)=10011$ $e(11)=11000$

26.

$$\begin{pmatrix} 0&0&0 \\ 0&0&1 \\ 0&1&0 \\ 0&1&1 \\ 1&0&0 \\ 1&0&1 \\ 1&1&0 \\ 1&1&1 \end{pmatrix} \begin{pmatrix} 1&0&0&1&0&0 \\ 0&1&0&0&1&1 \\ 0&0&1&1&1&1 \end{pmatrix} = \begin{pmatrix} 0&0&0&0&0&0 \\ 0&0&1&1&1&1 \\ 0&1&0&0&1&1 \\ 0&1&1&1&0&0 \\ 1&0&0&1&0&0 \\ 1&0&1&0&1&1 \\ 1&1&0&1&1&1 \\ 1&1&1&0&0&0 \end{pmatrix}$$

23. Let

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Decode the following words relative to a maximum likelihood decoding function associated with e_H.

(a) 10100 (b) 01101 (c) 11011

11.2 ⑪

P421-23.

① $e: B^2 \longrightarrow B^5$

$$\begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

②

00000	01101	10011	11110
00001	01100	10010	11111
00010	01111	10001	11100
00100	01001	10100	11010
01000	00101	11011	10110
10000	11101	00011	01110
00110	01011	10101	11000
10100	11001	00111	01010

③ $d(10100) = 00$

$d(01101) = 01$

$d(11011) = d(10011) = 10$

④ 注意. 译码表不唯一. 答案不唯一.

如 倒数第二行. 陪集头可为 11000

最后一行 陪集头可为 01010